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Continuous Behavioral Authentication for Mobile Devices Using an Adaptive LSTM-Based Intelligent Agent

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Abstract

Continuous authentication from smartphone inertial sensors must remain useful when a legitimate user's score distribution changes, yet an adaptive decision rule can also increase false acceptance or become vulnerable to profile poisoning. We evaluate a two-layer long short-term memory (LSTM) classifier and a lightweight agent that updates the first two exponential moving moments of accepted confidence scores. The study uses accelerometer and gyroscope data from 100 HMOG subjects. To prevent leakage, the revised protocol reads only the physical sensor columns, constructs non-overlapping sequences within individual sessions, keeps complete sessions disjoint between model fitting, validation, and test, and estimates normalization and initial operating points without test data. Linear SVM, random forest, and LSTM receive the same 10-second input. On clean held-out sessions, their macro equal error rates are respectively 29.92% ± 0.80%, 24.05% ± 0.86%, and 21.87% ± 1.01%. At the enrollment-derived operating point, the frozen LSTM threshold and the EMA agent obtain HTERs of 23.19% ± 0.95% and 24.53% ± 1.28% in a balanced alternating zero-effort attack stress test. Under a standardized synthetic drift of $\|\delta\|_2 = 2$, their HTERs are 24.89% ± 1.37% and 26.39% ± 2.12%. Thus the tested EMA rule does not improve the frozen threshold. These results provide a direct ablation of the adaptive component and delimit, rather than assume, its benefit. The work's primary contribution is a reproducible, session-disjoint evaluation of online score adaptation and an explicit analysis of its security–usability trade-off.

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Keywords: continuous authentication; behavioral biometrics; LSTM; adaptive threshold; inertial sensors; HMOG; concept drift

1. Introduction

One-time authentication cannot protect an unlocked smartphone after control of the device changes. Continuous authentication (CA) instead evaluates behavioral signals throughout a session, often without explicit interaction

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[1, 2]. Accelerometers and gyroscopes are attractive for this purpose because they are widely available and capture user-specific handling and motion patterns [3, 4].

The difficulty is temporal change. A classifier trained during enrollment can become poorly calibrated as posture, task, fatigue, or context changes [5]. Retraining a deep model after every observation is impractical on a mobile device, whereas a score-level update is inexpensive. That convenience is not evidence of effectiveness: an update rule must improve false rejection without opening the operating point to impostors, and it must be evaluated causally rather than by selecting a second threshold on test data.

We isolate the adaptation claim with a session-disjoint nested protocol, an explicit EMA decision equation whose parameters are fixed before testing, and a controlled frozen-LSTM versus LSTM+agent ablation at the same initial operating point. We report causal FAR, FRR, and HTER decisions on clean sessions and under synthetic drift, while limiting conclusions to the tested protocol.

2. Related Work and Scope of Novelty

CA has a substantial literature. Broad surveys cover system components, modalities, metrics, privacy, and open deployment problems [6, 7, 2]. A contemporary smartphone-focused survey catalogs more than 140 sensor and behavioral-biometric approaches [4]. The present work therefore does not claim that continuous authentication, inertial sensing, LSTM classification, or EMA by itself is new.

HMOG introduced hand-movement, orientation, and grasp features for smartphone authentication [3]. Subsequent work applied augmentation, deep sensor models, and adaptive verification to mobile behavioral data [8, 9, 5]. LSTM and hybrid recurrent networks are well established for sequential inertial biometrics [10, 11, 12]. Recent work has also studied smartphone authentication from gyroscope and accelerometer data for mobile educational resources [13]. In adjacent security analytics, graph-based typology features have been investigated for machine-learning fraud detection [14], alongside graph preprocessing and analytical pipelines for financial-crime detection [15, 16]. Other applied-ML studies by the authors address cataract detection, UAV recognition, and IoT energy management [17, 18, 19]; these are author-context references rather than direct CA benchmarks. Exponentially weighted statistics are likewise standard tools for tracking non-stationary streams [20, 21].

Table 1. Positioning of the present evaluation. A check mark means the cited work explicitly contains the listed element; protocols and datasets differ, so reported error rates are not directly comparable.

Work	Inertial	Sequence model	Online update	Drift test
Sitová et al. [3]	✓	–	–	–
AUToSen [9]	✓	✓	–	–
Verma et al. [5]	✓	–	✓	✓
This study	✓	✓	✓	✓

The novelty lies in the controlled combination: frozen and EMA-updated LSTM thresholds start from the same enrollment profile and face identical chronological test scores, isolating adaptation from classifier performance.

3. Dataset and Leakage-Resistant Protocol

3.1. HMOG data and parsing

HMOG contains 24 smartphone-interaction sessions for each of 100 participants, including sitting and walking tasks [3]. We use the accelerometer and gyroscope. An HMOG inertial CSV row contains metadata before the three physical axes; consequently, only columns 4–6 are read as x, y, z. This detail is security-critical because a session identifier must never enter the feature vector. Six empty sensor files belonging to one subject are logged and skipped; that subject retains enough valid sessions to appear in every partition.

After a five-second offset, the first 20 non-overlapping windows of 100 samples are retained from each valid session. Each window is represented by mean, standard deviation, minimum, maximum, root mean square, and skewness for

each of six channels, producing 36 features. Ten adjacent windows form a 10-second sequence. The sequence step is also ten windows, so evaluation endpoints do not share raw samples. Sequences never cross a session boundary.

3.2. Nested session split

For session number r, define $g(r) = (r - 1) \bmod 5$. In outer fold f, sessions with $g(r) = f$ are test sessions, sessions with $g(r) = (f + 1) \bmod 5$ are validation/enrollment sessions, and the remaining sessions are used to fit the model. Whole sessions, rather than sequences, are assigned to partitions. Every included subject is required to occur in all three partitions. This is a session-disjoint cross-validation protocol, not a claim of leave-one-subject-out or prospective unseen-subject recognition.

A standard scaler is fitted only to the model-fitting sequences and then applied unchanged to validation and test sequences. Early stopping and initial threshold estimation use validation data; test labels are not used for either.

4. Method

4.1. System overview

The system has five stages:

$$\boxed{\text{sensor samples}} \rightarrow \boxed{\text{1-s windows}} \rightarrow \boxed{\text{36 features}} \rightarrow \boxed{\text{LSTM score}} \rightarrow \boxed{\text{static or EMA decision}}$$

The first four stages are identical in the ablation. Only the last-stage threshold update changes, which makes any difference in FAR/FRR attributable to the agent rather than to a different classifier or observation window.

4.2. LSTM classifier and baselines

The input is $\mathbf{X} \in \mathbb{R}^{10\times 36}$. A two-layer LSTM with 64 hidden units per layer returns its last hidden state. Layer normalization, a 32-unit ReLU layer, dropout of 0.3, and a 100-class softmax produce claimant scores $s_{u,n} \in [0, 1]$. Training uses cross-entropy, Adam with learning rate 10^{-3}, batch size 256, a maximum of 20 epochs, and early-stopping patience 4.

Linear SVM and random forest baselines receive the same sequence flattened to 720 dimensions. The forest contains 300 trees and has minimum leaf size 2. These baselines compare classifier families at the same temporal endpoint; they are not presented as adaptive systems.

4.3. EMA adaptive agent

For claimant u, the validation genuine scores initialize the first and second moments $p_{u,0}$ and $q_{u,0}$. Before attempt n, the agent computes

$$\sigma_{u,n-1} = \sqrt{\max(q_{u,n-1} - p_{u,n-1}^2, \epsilon^2)}, \tag{1}$$

$$\tau_{u,n} = \max(0, p_{u,n-1} - k\sigma_{u,n-1}), \tag{2}$$

$$d_{u,n} = \mathbb{I}[s_{u,n} \geq \tau_{u,n}]. \tag{3}$$

When $d_{u,n} = 1$, the accepted score updates both moments:

$$p_{u,n} = (1 - \alpha)p_{u,n-1} + \alpha s_{u,n}, \tag{4}$$

$$q_{u,n} = (1 - \alpha)q_{u,n-1} + \alpha s_{u,n}^2. \tag{5}$$

After a rejection the state is unchanged. Thus repeated unauthenticated events cannot lower the threshold. An accepted impostor can still poison the profile; the evaluation deliberately preserves that risk instead of using ground-truth labels to gate updates.

We fix $\alpha = 0.05$, corresponding to an effective memory of approximately $1/\alpha = 20$ accepted attempts. For each claimant, the validation genuine and impostor scores determine an EER threshold $\tau_{u,0}$; the corresponding $k_u = (p_{u,0} - \tau_{u,0})/\sigma_{u,0}$ makes the first adaptive threshold exactly equal to that independently calibrated operating point. Thus k_u is not selected on test scores. The frozen LSTM ablation uses the same $\tau_{u,0}$ but never updates it. There is no separate sliding mean and no unexplained $K = 10$ score buffer.

5. Evaluation

For each claimant, equal error rate (EER) is found from all exact ROC thresholds on held-out scores, then macro-averaged across claimants and folds. This EER is a score-separability statistic; its test-derived crossing threshold is not used for online decisions.

The static and adaptive decisions are evaluated at their enrollment-derived operating points. Genuine test attempts follow session and within-session chronology. Each is alternated with a zero-effort impostor score sampled without replacement; sampling is repeated ten times. This balanced 50% attack stream is an explicit stress test, not an estimate of real attack prevalence. Because the agent sees no labels, an accepted impostor updates its profile. We report macro false acceptance rate (FAR), false rejection rate (FRR), and HTER = (FAR + FRR)/2.

To isolate score drift, a deterministic direction in standardized 36-feature space is added progressively over each subject's test chronology. Magnitudes $\|\delta\|_2 \in \{0, 1, 2\}$ are tested on the same trained model. This is a controlled covariate-shift experiment; it does not claim to reproduce every physiological form of natural behavioral drift.

6. Results

Table 2. Clean held-out-session macro EER (mean ± standard deviation over five outer folds). All methods use the same 10-second sequences.

Model	EER (%)
Linear SVM	29.92 ± 0.80
Random forest	24.05 ± 0.86
LSTM (raw score)	21.87 ± 1.01

Table 2 compares score separability. Table 3 isolates the agent at a threshold estimated without test data. The agent is not assigned an additional post-hoc EER; its actual causal decisions are reported.

Table 3. Frozen-threshold and EMA-agent operating points in the balanced alternating stream (mean ± standard deviation over five folds).

$\|\delta\|_2$	Decision rule	FAR (%)	FRR (%)	HTER (%)
0	Frozen	22.55 ± 1.13	23.82 ± 2.27	23.19 ± 0.95
0	EMA agent	24.46 ± 2.12	24.60 ± 2.04	24.53 ± 1.28
1	Frozen	22.43 ± 0.69	24.82 ± 1.97	23.63 ± 1.10
1	EMA agent	24.30 ± 2.00	25.73 ± 2.64	25.02 ± 1.66
2	Frozen	22.40 ± 0.85	27.37 ± 2.91	24.89 ± 1.37
2	EMA agent	24.53 ± 2.11	28.25 ± 4.25	26.39 ± 2.12

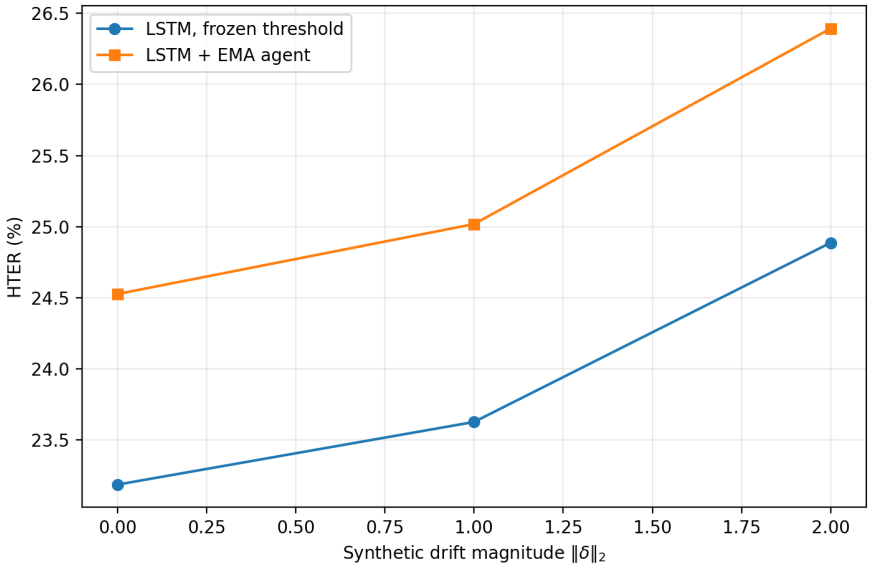


Fig. 1. HTER of the frozen LSTM threshold and EMA agent under synthetic standardized-feature drift. Error bars are omitted because fold-level values are reported in the accompanying result file.

On clean data, the LSTM lowers macro EER by 2.18 percentage points relative to RF and by 8.05 points relative to the linear SVM.

The agent ablation is negative. On clean sessions, adaptation increases HTER by 1.34 points. At drift magnitudes 1 and 2, it increases HTER by 1.39 and 1.51 points, respectively; these differences are computed from the unrounded fold means. The frozen threshold has lower HTER in every one of the five folds at all three drift levels. Adaptation increases both FAR and FRR on average; therefore its poorer HTER cannot be characterized as a deliberate security–usability trade. Both decision rules degrade as drift grows, and the tested EMA mechanism does not preserve the operating point better than the static comparator.

7. Discussion

Table 2 measures score discrimination, Table 3 tests an independently initialized operating point, and Fig. 1 isolates the effect of EMA adaptation under controlled shift. Low EER alone does not establish an agent contribution, nor is lower FRR beneficial when FAR rises enough to worsen HTER.

The causal accepted-only update cannot lower a threshold after rejection, but also cannot adapt to sub-threshold genuine scores and remains vulnerable to poisoning after false acceptance. Deployment would require high-confidence update gates, step-up authentication, and explicit poisoning tests.

7.1. Limitations

Evaluation is limited to HMOG and cannot establish generalization across devices, populations, or corpora. Six empty sessions are skipped. The modulo-session split prevents overlap but is not a rolling-origin forecast; the 50% impostor stream is a stress test, not a prevalence model. Synthetic drift uses one feature-space direction per fold. Future work requires another session-rich dataset, natural and varied sensor shifts, session-clustered intervals, and on-device latency and energy measurements.

Published error rates are not ranked because activities, modalities, enrollment, subjects, and split rules differ.

8. Conclusion

The leakage-resistant, session-disjoint HMOG evaluation separates validation from training and compares the EMA agent with the same LSTM under a frozen threshold. Although the LSTM has the best clean EER among the classifiers, EMA adaptation consistently worsens HTER and should not be presented as a successful drift solution. Broader claims require independent-dataset and live-stream validation. The companion code records parsing, folds, skipped sessions, seed, hyperparameters, and fold-level results; its canonical entry point is `hmog_revised_experiment.py`.

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